

PROJECT PHASE II - REVIEW 1

AI-POWERED HEALTH REMINDER SYSTEM AND EMERGENCY ALERT APP

TEAM MEMBERS

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PROBLEM STATEMENT

- **Health Monitoring Challenges:**

Elderly and chronically ill individuals require constant assistance.

- **Issues With Traditional Reminder Systems:**

Provide static reminders without intelligent tracking or real-time monitoring.

- **Risk of Undetected Emergencies :**

Missed medication and prolonged inactivity often go unnoticed.

- **Lack of User Friendly Features :**

Most apps are not designed with elderly accessibility.

MOTIVATION

- Elderly and sick people may forget to take medicines on time.
- Missing medicines or help during emergencies can be dangerous.
- Normal reminder apps are not smart and do not detect emergencies.
- Many apps are hard for elderly people to use.
- There is a need for a simple and smart health app.
- Using AI and sensors can help detect emergencies quick.

OBJECTIVES

- Enhance independence and safety for elderly and chronically ill individuals.
- Provide a user-friendly mobile app with intelligent medicine reminders.
- Enable real-time emergency detection using sensors (accelerometer).
- Facilitate caregiver support through a live monitoring dashboard.
- Incorporate voice commands for hands-free navigation.
- Use facial recognition for alert cancellation to ensure safety.
- Automate appointment reminders and manage healthcare provider visits.
- Ensure timely medication adherence and track missed doses.
- Monitor medicine stock .

KEY PAPERS

1. Intelligent and Safe Medication Box In Health IoT Platform for Medication Monitoring System with Timely Reminders.
2. MEMO Box: Health Assistant for Depression with Medicine Carrier and Exercise Adjustment Driven by Edge AI .
3. IoT Based Medical Assistant for Efficient Monitoring of Patients in Response to COVID-19.
4. ACTIVA: Innovation in Quality of Care for Nursing Homes through Activity Recognition.
5. Voice Assistants for Health Self-Management: Designing for and with Older Adults.

RESEARCH GAP

1. Intelligent and Safe Medication Box In Health IoT Platform for Medication Monitoring System with Timely Reminders.

- Lacks emergency detection (inactivity).
- No voice/multilingual support.
- No facial recognition Limited caregiver dashboard.
- No doctor/appointment reminders.
- Incomplete stock tracking.
- Limited elderly accessibility.

2. MEMO Box: Health Assistant for Depression with Medicine Carrier and Exercise Adjustment Driven by Edge AI .

- No caregiver dashboard, voice navigation.
- Lacks real-time emergency detection (falls, SOS).
- Focuses on depression management, not general elderly care.

3. IoT Based Medical Assistant for Efficient Monitoring of Patients in Response to COVID-19.

- Hardware-heavy (sensors + microcontrollers).
- No caregiver dashboard or mobile family alerts.
- No accessibility features (multilingual, SOS shake/face cancel).
- Limited to COVID, not broader elderly/chronic carebody text.

4. ACTIVA: Innovation in Quality of Care for Nursing Homes through Activity Recognition.

- No medicine reminders.
- No AI predictions (just rules).
- No voice or multilingual support.
- No face-based alert cancellation.
- Caregiver focus only, not family.

5. Voice Assistants for Health Self-Management: Designing for and with Older Adults.

- No emergency alert system.
- No stock tracking or doctor scheduling.
- No face-based alert cancellation.
- Only for elderly, not all ages.
- No family caregiver dashboard.

SUMMARY TABLE

<i>REF</i>	<i>TITLE</i>	<i>KEY FEATURES</i>	<i>GAPS IDENTIFIED</i>
[1]	Intelligent and Safe Medication Box In Health IoT Platform for Medication Monitoring System with Timely Reminders - IEEE, 2021	• Smart medication box with 6 sub-boxes for organizing pills. • Timely reminders via Android app for time, dosage, and stock. • Bio-sensors for temperature and heartbeat monitoring. • Authentication to prevent misuse/overdosage.	• Lacks emergency detection (falls, inactivity). • No voice/multilingual support. • No facial recognition • Limited caregiver dashboard. • No doctor/appointment reminders. • Incomplete stock tracking. • Limited elderly accessibility.
[2]	MEMO Box: Health Assistant for Depression with Medicine Carrier and Exercise Adjustment Driven by Edge AI – IEEE, 2023	• Smart medicine box with medication management. • Emotion recognition using AI & EEG data. • Exercise recommendation through reinforcement learning. • Edge & cloud integration for low latency response.	• Focuses on depression management, not general elderly care. • Requires dedicated hardware (box, wearables). • No caregiver dashboard, multilingual voice navigation. • Lacks real-time emergency detection (falls, SOS).

<i>REF</i>	<i>TITLE</i>	<i>KEY FEATURES</i>	<i>GAPS IDENTIFIED</i>
[3]	IoT Based Medical Assistant for Efficient Monitoring of Patients in Response to COVID-19 – IEEE, 2020	• Biomedical sensors: heartbeat, temperature, motion, saline level. • Data auto-uploaded to cloud (OneDrive) every 30s. • Automated alarms for abnormal readings. • Android app for bed control, reminders, nurse call. • Enables remote monitoring, reduces infection risk.	• Designed mainly for hospital COVID-19 monitoring. • Hardware-heavy (sensors + microcontrollers). • No caregiver dashboard or mobile family alerts. • No accessibility features (multilingual, SOS shake/face cancel). • Limited to COVID, not broader elderly/chronic care.
[4]	S. González-Mendoza et al., “ACTIVA: Innovation in Quality of Care for Nursing Homes through Activity Recognition,” IEEE Access 2023.	• Uses HAR (activity recognition) for elderly. • Sensors to detect activities. • Sends real-time alerts to caregivers.	• Works only in nursing homes. • No medicine reminders. • No AI predictions (just rules). • No voice or multilingual support. • No face-based alert cancellation. • Caregiver focus only, not family.

<i>REF</i>	<i>TITLE</i>	<i>KEY FEATURES</i>	<i>GAPS IDENTIFIED</i>
[5]	M. Sezgin et al., "Voice Assistants for Health Self-Management: Designing for and with Older Adults," Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems, Hamburg, Germany, Apr. 2023.	• Voice assistant for elderly care. • Medicine reminders & health info. • Easy language interaction.	• Only voice, no multimodal (face/UI). • No emergency alert system. • No stock tracking or doctor scheduling. • No face-based alert cancellation. • Only for elderly, not all ages. • No family caregiver dashboard.
[6]	H. Tian et al., "Health Monitoring and Automatic Notification Device Based on Laser-Induced Graphene", IEEE Sensors Journal, Jul. 2019.	• LIG sensors for respiration monitoring. • Detects apnea & abnormal breathing. • Sends automatic health alerts.	• Only respiration monitoring • No medicine reminders • No voice or multilingual support • No face detection alert cancel • No doctor/medicine management • Hardware-only focus

PROPOSED SOLUTION

Methodology

1. User Data Collection

- Collects essential details such as medicine information, doctor appointments, and caregiver contacts.
- Captures user interactions including button presses, voice commands, and SOS actions through the app.

2.Doctor Scheduling & Appointment Reminder Module

- Allows users to schedule and manage doctor appointments.
- Sends reminder notifications and maintains appointment history.

3.Medicine Reminder Module

- Sends timely alarms and notifications for medicine intake.
- Records missed or delayed doses for caregiver monitoring.

4. Emergency Alert Module

- Enables users to trigger SOS manually using a button or gesture.
- Immediately sends emergency alerts to the backend system.

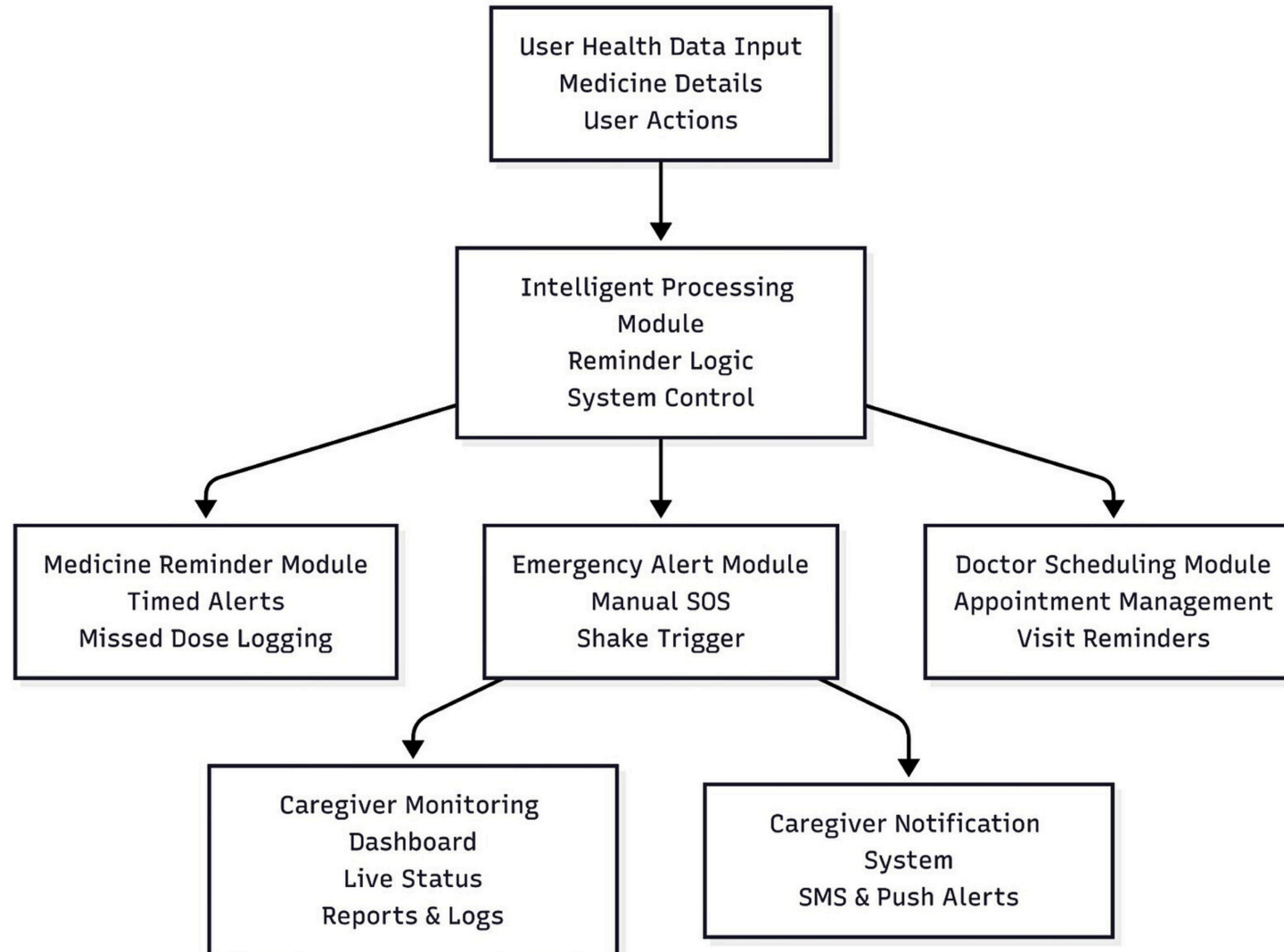
5. Caregiver Notification & Monitoring

- Sends SMS and push notifications during emergencies or missed medicines.
- Allows caregivers to monitor user status through reports and dashboards.

6. Voice Navigation Module

- Supports voice commands.
- Enables elderly-friendly, hands-free navigation.
- CNN language is used for voice navigation.

PROPOSED SOLUTION



NOVELTY & SCOPE

Innovation

- AI-based medicine reminder and emergency detection system.
- Uses smartphone sensors for inactivity detection.
- Face recognition to cancel false emergency alerts.
- Voice-enabled elderly-friendly interface.

Implementation Scope

- Mobile app for elderly and chronically ill users.
- AI logic for reminders and emergency alerts.
- Cloud backend for data storage and notifications.
- Caregiver dashboard for remote monitoring.

Advantages

- Improves user safety and independence.
- Reduces missed medicines and false alerts.
- Easy to use with voice support.
- Cost-effective using existing smartphone hardware.

TASK DISTRIBUTION

Team member roles

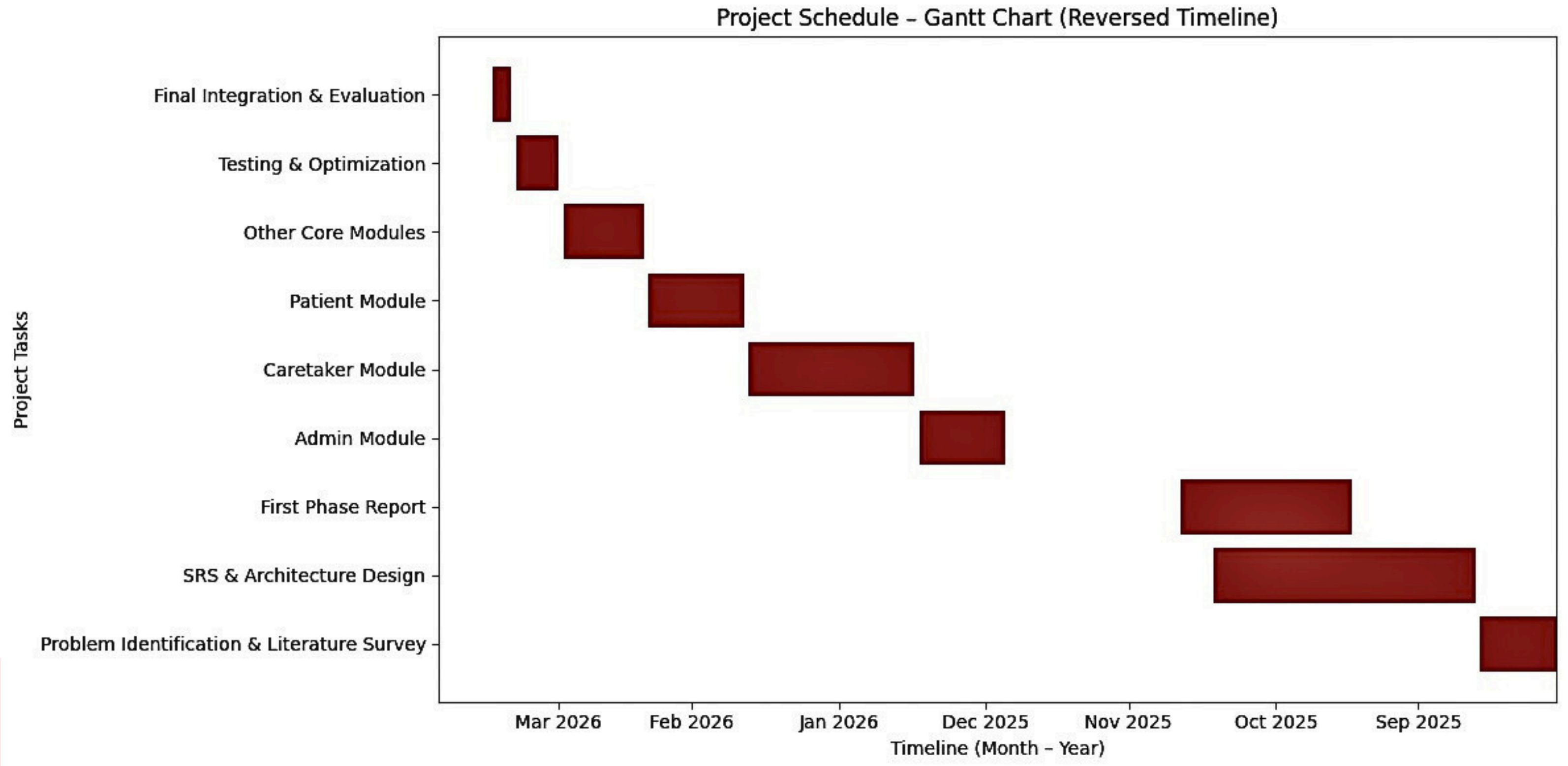
1. Sreelakshmi Sai - Login , Dashboard & Emergency Detection
2. Jyothylakshmi Biju - Patient , Caregiver Data & Voice Navigation
3. Feba M Aji - Mobile Application & Doctor Appointment
4. Shibana Nazeer - AI-chatbot & Face Detection

Work Breakdown

- Login , Dashboard & Emergency Detection
 - > Created login page
 - > User roles (Caregiver, Family, Medical staff)
 - > Dashboard screen design
 - > Alert system and cancellation

- Patient , Caregiver Data & Voice Navigation
 - > Added patient details
 - > Caregiver management
 - > View, add, and delete patients
 - > Used for spoken commands
- Mobile Application & Doctor Appointment
 - > Patient registration in mobile & appointments of the doctor
 - > Manage patient's screen
 - > Medicine button and alerts
 - > Manage and simplify daily activities
- AI Chatbot & Face Detection
 - > Provides daily details of the patient
 - > Cancel alert
 - > Find emotions and continues monitoring

PROJECT SCHEDULE



Milestones

- > Problem identification and literature survey - Completed
- > SRS and architecture design - Completed
- > First phase report - Completed
- > Admin Module - Completed
- > Caretaker Module - Completed
- > Patient Module - Completed
- > Other core module implementations - Completed
- > Testing and optimization - Completed
- > Final integration and evaluation - Completed

INTERIM RESULTS

Current Progress

- Admin module implemented
- Mobile app UI development implemented.
- Medicine reminder and scheduling module implemented.
- Sensor-based inactivity detection implemented.
- Voice commands implemented.
- Backend and caregiver dashboard implemented.
- Emergency alert integration implimented.

TECHNOLOGY STACK

- **Operating System:** Microsoft Windows 8 or Above
- **Front End:** HTML, CSS, JavaScript, Flutter
- **Back End:** Python
- **Database:** MySQL Server
- **IDE / Development Tools:** VS Code, Wamp / Xampp, SqlYog
- **AI Model:** Gemini api

IMPLEMENTATION

User Registration and Authentication

- Users register and log in through the application interface.
- The frontend is developed using Flutter, HTML, CSS, and JavaScript.
- The Django backend manages authentication and user validation.
- User data is stored securely in the MySQL database.

Medicine Reminder Module

- Users can add and manage their medication schedules.
- Reminder details are sent to the Django backend through HTTP requests.
- The medication data is stored in the MySQL database.
- The system sends notifications at the scheduled reminder time.

AI Chatbot Module

- The system integrates Google Gemini to provide chatbot support.
- Users can ask health-related questions through the chatbot.
- The chatbot processes queries using Natural Language Processing (NLP).
- It generates helpful responses for users.

Emergency Alert Module

- The system uses the accelerometer sensor to detect sudden falls or abnormal movements.
- Sensor data is continuously monitored by the mobile application.
- If a fall or emergency is detected, the system automatically triggers an alert.
- A call is sent to the registered caregiver for immediate assistance.
- This ensures quick response during emergency situations.

Voice Notification Module

- The application uses Text-to-Speech (TTS) for voice reminders.
- Reminder notifications are converted into voice messages.
- Speech-to-Text (STT) allows users to give voice commands.
- This improves accessibility for elderly users.

Emotion Detection Module

- The system detects user emotions using facial expression analysis.
- The camera captures the user's facial image.
- The backend processes the image using an emotion detection model.
- The system identifies emotions such as happy, sad, or angry.

Screenshots



Welcome to CareTrack

Sign in to access the patient monitoring dashboard


Caregiver


Family Member


Medical Staff

Email Address

 admin@gmail.com

Password

 Enter your password 

☐ Remember me [Forgot password?](#)

Sign In

OR

Real-time Health Monitoring

Track vital signs, medication schedules, and activity patterns with our innovative system designed specifically for dementia care.

Monitor Health Data



Patient Management

**Rajesh D**
Professional Caregiver - FamilyMember object (3)

ID:	6
Phone:	7896852356
Email:	rajesh@gmail.coml
Date of Birth:	22/3/1973
Gender:	Male

System Features

-  Health Monitoring
-  Location Tracking



Welcome to MindCare Connect

Dementia Care Management System



Healthcare Login

Username or Email

Password

Enter your credentials to access dementia care management

SIGN IN

New to MindCare?

CAREGIVER

Caregiver Registration

Register as a caregiver to provide support and manage dementia care

Full Name *

Email Address *

Phone Number *

Gender *

Male

Address *

Enter your complete address

Phone Number *

Gender *

Male

Address *

Phone Number *

Gender *

Male

Address *



MindCare Connect

Dementia Care Management System

Connect to your healthcare server to access patient care tools



Connected Successfully!

Ready to access dementia care management tools

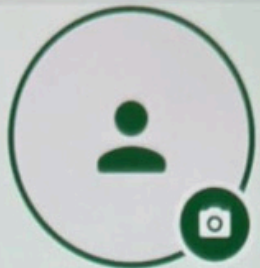
ACCESS PATIENT CARE

CONNECT TO HEALTHCARE

Need help finding your server IP?

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← Patient Registration



Add Profile Photo

Dementia Care Patient Registration

Create your healthcare account to access personalized care

 Full Name *

 Email Address *

 Phone Number *

 Date of Birth *

Gender *

 Male

← My Profile



Manvi
Daughter
Age: 22 years

 Email
manvi@gmail.com

 Phone
8956253685

 Date of Birth
15/2/2003
Age: 22 years

 Gender
Female

 Address
ABCD

Missed Medicines

 **Fibroadenosis** 
Missed 1 times

Close

Medicine Assistant



മരുന്ന് അസിസ്റ്റന്റ് തയ്യാറാണ്. മരുന്ന് സ്ഥിരീകരിക്കാൻ ശബ്ദം റെക്കോർഡ് ചെയ്യുക.

 Record Voice



Edit Appointment

Edit Appointment

For Patient: Manvi

Appointment Information

Doctor Name *

Dr.Vinayek R

Specialization *

Gynecologist

Hospital/Clinic Name *

KM hospital

Appointment Date *

2026-03-09

Appointment Time *

16:00

Contact Phone *

9632586853

Mind...

Good Evening,
Manvi!

Monday, March 9

Your health is our priority

1

Today's
Medicines

1

Upcoming
Appointments

1

Missed
Medicines

Emotion Monitoring

Emergency Check

Status

Last capture: 21:15:22

Captures: 16 images

Stop Monitoring

MindCare Conn...

Emergency Features

Status:

Last Activity: 12:42:37

Contacts: 1

Threshold: 30 min

Last Alert: INACTIVITY - 12:42:54

Emergency Face Check in Progress

Checking if you're okay before sending alert

Shake to Alert

Shake phone to send emergency alert (with face check)

Inactivity Monitoring

Alert if phone inactive for 30+ minutes (with face check)

Before sending alerts, face emotion will be checked. If no face detected within 25 seconds, alert will be sent.

Test Alert

Simulate Activity

Medicine Reminders

Upcoming Appointments

Dr. Dr.Vinayek R

KM hospital

Mar 9 16:00

Visited

NOT Visited

Dementia Care Tips

Establish a daily routine for consistency

Use reminders and notes for important tasks

Stay physically active with gentle exercises

Engage in mentally stimulating activities

Maintain social connections and interactions

Emergency Contact

Need immediate help? Contact your caregiver or emergency services

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Emergency Contacts

Manvi

*

Add Emergency Contact

For Manvi

Contact Information

Full Name *

Phone Number *

Relationship Details

Relationship to Patient *

Spouse

How is this person related to the patient?

This contact will be notified during emergencies and critical situations

CANCEL

+ ADD

Emergency Contacts

Manvi

*

Jyothylakshmi Biju

Sibling

8129619554

Edit

Delete

<

331

S

Sreeee >

Text Message • SMS

Today, 12:42

INACTIVITY ALERT! Manvi's phone has been inactive for 30 minutes. Last activity: 12:42:37. Please check on them. Location: Lat: 9.9509, Long: 76.6318

Emergency Alert Sent

Inactivity alert has been sent to your family members.

Time: 12:42:54

OK

Emergency Face Click

* Emergency Calls

Calling emergency contacts...

No response for Fibroadenosis after 3 reminders

Cancel

CHALLENGES & RISK

Technical Challenges

- Detecting Emergencies Correctly
- Sensor Accuracy
- App Performance
- Internet Dependency
- Data Storage and Security

APPLICATIONS

- Elderly Care Management – Helps elderly people remember medicines and send alerts.
- Chronic Disease Monitoring – Supports patients with long-term diseases.
- Remote Caregiver Monitoring – Allows caregivers to monitor users remotely.
- Home Healthcare Support – Assists patients in home healthcare settings.
- Hospital and Clinic Support – Helps patients follow medication schedules.
- Emergency Health Assistance – Sends SOS alerts during emergencies.
- Health Management Systems – Can be integrated with digital health platforms.

LIMITATIONS

- Application performance depends on mobile battery life.
- Limited real-time health monitoring sensors in the current system.
- Requires user permission for camera, sensors, and notifications.
- Caregiver monitoring depends on internet availability.
- Limited personalization for different medical conditions.

FUTURE WORK

- Add GPS-based emergency location sharing.
- Integrate hospital and ambulance services for faster response.
- Provide detailed health analytics and reports for caregivers.
- Improve voice assistant accuracy and multilingual support.
- Enable cloud-based data backup and security improvements.
- Add AI-based medicine dosage recommendations (with doctor support).

CONCLUSION

Summary.

- AI-powered system for health reminders and emergency alerts
- Improves medication adherence and user safety
- Enables real-time caregiver notifications
- User-friendly design focused on elderly needs
- Scalable for future healthcare enhancements

PUBLICATIONS & IPR



The background features two large, light pink geometric shapes. One is a triangle on the left side, and the other is a quadrilateral on the right side, both pointing towards the center.

THANK YOU